

Ahmed Radwan

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TECHNICAL SKILLS

Languages: Python, SQL, Java

Python Packages: Hugging Face Transformers, LangChain, LangGraph, OpenAI API, vLLM, Accelerate, Whisper, PyTorch, TensorFlow, Scikit-learn, PySpark, Pandas, NumPy, Matplotlib

Concepts: RAG, Embeddings, Vector Search, Prompt Engineering, Instruction Tuning, Fine-Tuning (LoRA), Experiment Tracking, Model Evaluation, Distributed Training, Quantization, Model Compression, Self-Supervised Learning

Tools: Git, Singularity, Docker, Linux

EDUCATION

Master of Science - Computer Science

York University

3.96/4.0 GPA.

Toronto, Canada

PROFESSIONAL EXPERIENCE

Associate Applied Machine Learning Specialist

Vector Institute

09/2025 – Present

Toronto, Canada

- Pioneered **SONIC-O1**, the first omnimodal benchmark for audio-video MLLMs (231 videos, 5k human-verified QAs); identified a 22.9% temporal grounding gap in Gemini 3.0 Pro vs. open-source models.
- Architected **SONIC-O1-Agent**, a multi-agent framework for evidence-based video understanding; utilized cross-modal grounding and tool-orchestration to deliver interpretable reasoning for complex long-context multimodal tasks.
- Developed and shipped **UnBias+**, a Python package for bias detection using instruction-tuned Qwen-3 models with character-level offset prediction and a FastAPI serving layer.
- Optimized large-scale inference by **cutting GPU costs 50% using tensor-parallel sharding** (vLLM + Accelerate) on A40 clusters; authored industry-standard fairness guidance aligned with EU AI Act and NIST RMF.

Research Assistant

York University

08/2024 – 04/2026

Toronto, Canada

- Achieved **92% SOTA accuracy** for Wi-Fi activity recognition via self-supervised contrastive learning, successfully eliminating the need for labeled data across multi-domain benchmark scenarios.
- Improved **inference 17x with 20% smaller models** by compressing time-series representations, maintaining 90%+ fidelity for edge deployment.
- **Reduced 90% of labeled data needs** via few-shot meta-learning, adapting during deployment with fewer than 100 examples.
- **Developed MU-SHOT-Fi**, boosting multi-user Wi-Fi activity recognition by 22% under domain shifts without labeled source data.

Research Engineer

Asas.Ai

09/2023 – 06/2025

- Developed a Context-Aware Story generation framework using Large Language Models (LLMs), integrating multimodal input via the GPT-4 Vision API.
- Enhanced language understanding and task-specific performance through instruction-tuning on Arabic datasets for various creative writing tasks.

Research Engineer

King Abdullah University of Science and Technology (KAUST)

02/2024 – 10/2024

- **Reduced memory by 75%** while maintaining 90%+ accuracy using TinyML NLP quantization + compression for edge devices.
- Achieved **30% efficiency gain over federated learning** by implementing Split Learning for privacy-preserving NLP; measured efficiency and CO2-aware tradeoffs.

Artificial Intelligence Intern

King Abdullah University of Science and Technology (KAUST)

07/2023 – 08/2023

- **Selected among the top 100 out of 10,000** applicants for an elite, fully funded AI program.
- Built and optimized generative models for deep unsupervised learning, focusing on scalable real-world applications.

- Applied advanced NLP methods for sentiment analysis and machine translation across multilingual datasets.

PROJECTS

Multi-agent Framework for Finding Audio-Visual Needles by Reasoning over Multi-Video Haystacks

- Implemented the MAGNET framework to enhance Audio-Visual understanding and recommendation systems.
- Utilized transformers and open-source models implementations and documenting use cases.

Context-Aware Recommender for Fairness Requirements Engineering

- Implemented the ReFAIR framework for fairness-aware requirements engineering.
- Validated reproducibility and documented findings to support fairness research.

Fairness-Aware Medical Imaging

- Implemented adversarial and balanced fine-tuning methods to mitigate demographic bias in chest X-ray classification.
- Achieved improved fairness across race, gender, and insurance subgroups without sacrificing clinical performance.

PerfectPrayer

- Developed a real-time prayer motion tracking system using smartphone IMU sensors, deployed on Android devices.
- Applied time-series data preprocessing and motion recognition algorithms to for real-time accuracy during activities.

PUBLICATIONS

Contrastive Predictive Coding with Compression for Enhanced Channel State Feedback in Wireless Networks

IEEE Transactions on Neural Networks and Learning Systems (TNNLS)

Radwan, A. Y., Muhammad, F. S., Baker, M. & Tabassum, H.,

MU-SHOT-Fi: Self-Supervised Multi-User Wi-Fi Sensing with Source-free Unsupervised Domain Adaptation [↗](#)

IEEE Internet of Things (IoT)

Radwan, A. Y., Hina T.

SONIC-O1: A Real-World Benchmark for Evaluating Multimodal LLMs on Audio-Video Understanding [↗](#)

Preprint

Radwan, A. Y., Christos E., Hina T., Pandya, Deval, & Raza, Shaina.

Reducing Hallucinations in LLMs via Factuality-Aware Preference Learning [↗](#)

Association for Computational Linguistics (ACL 2026)

Chaduvula, S., Radwan, A. Y., Farooq, A., Ioannou, Y., & Raza, Shaina.

Transparency in Agentic AI: A Survey of Interpretability, Explainability, and Governance [↗](#)

Preprint

Raza, Shaina., Radwan, A. Y., Chaduvula, S., Alinoori, M., Christos E.

A Tutorial-cum-Survey on Self-Supervised Learning for Wi-Fi Sensing: Trends, Challenges, and Outlook [↗](#)

IEEE Communications Surveys and Tutorials

Radwan, A. Y., Mustafa Y., Navid H., Hina T., & Shahrokh V.

TinyML NLP Approach for Semantic Wireless Sentiment Classification [↗](#)

2025 EuCNC & 6G Summit - AI4C.

Radwan, A. Y., Shehab, M., & Alouini, M.S.

SARD: A Human-AI Collaborative Story Generation [↗](#)

HCI International 2024

Radwan, A. Y., Alasmari, K. M., Abdulbagi, O. A., & Alghamdi, E. A.

GRANTS & AWARDS

AIXPERT - EU Horizon Europe Research & Innovation Action

Grant Agreement No. 101214389, funded by the European Union, Jun 2025–May 2028

NSERC

Funded by the Natural Sciences and Engineering Research Council of Canada, Sep 2024–Apr 2026

1st Place Winner for 2024 Student Games [↗](#)

Organized By International Society of Automation and Sponsored by Aramco

1st Place Winner at Sehah Thon [↗](#)

Ministry of Health with Ministry of Hajj and Umrah